Mineral identification sheet (1/4)

Diamond

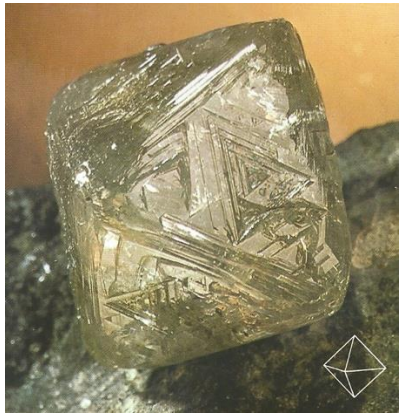
Formula: C

Color: colourless, yellowish to yellow, brown, black, blue, green or red, pink, champagne-tan, cognac-brown

Lustre: adamantine, greasy

Hardness: 10

Crystal System: Isometric



Diopside

Formula: $CaMgSi_2O_6$

Color: light to dark green, blue, colourless, snow white, grey, pale violet

Lustre: vitreous, dull

Hardness: 5.5-6.5

Crystal System: monoclinic





Garnet group

Formula: $(Mg,Ca,Fe,Mn)_3Al_2(SiO_4)_3$

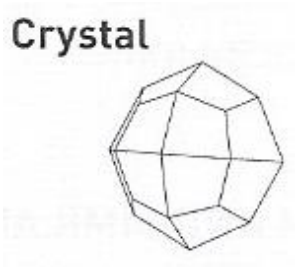
Color: <u>almandine</u> - Deep red, brownish red, red-violet, black; <u>pyrope</u> - Blood red, orange red, purple red, pink, black red

Lustre: vitreous, resinous

Hardness: 6.5-7

Crystal System: isometric





Mineral identification sheet (2/4)

Ilmenite

Formula: $FeTiO_3$

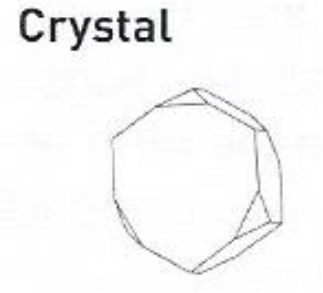
Color: iron black, black

Lustre: metallic

Hardness: 5-6

Crystal System: trigonal





K Feldspar

Formula: $(Na,K)AlSi_3O_8$

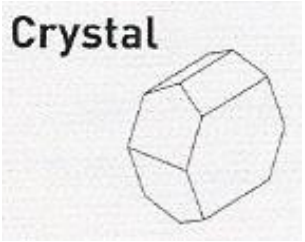
Color: Colourless, white, grey, yellowish white, or reddish white

Lustre: vitreous, nacreous

Hardness: 6

Crystal System: monoclinic/triclinic





Olivine

Formula: $(Mg,Fe)_2SiO_4$

Color: pale green, pale yellow, white

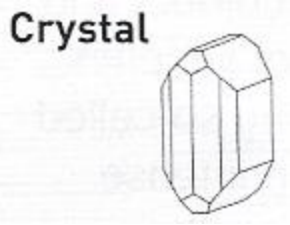
Lustre: vitreous

Hardness: 7

Crystal System: orthorhombic







Mineral identification sheet (3/4)

Quartz

Formula: SiO_2

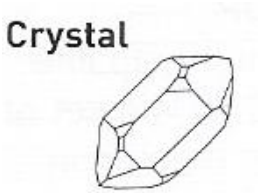
Color: Colorless, purple, rose, red, black, yellow, brown, green, blue, orange

Lustre: vitreous

Hardness: 7

Crystal System: trigonal





Calcite

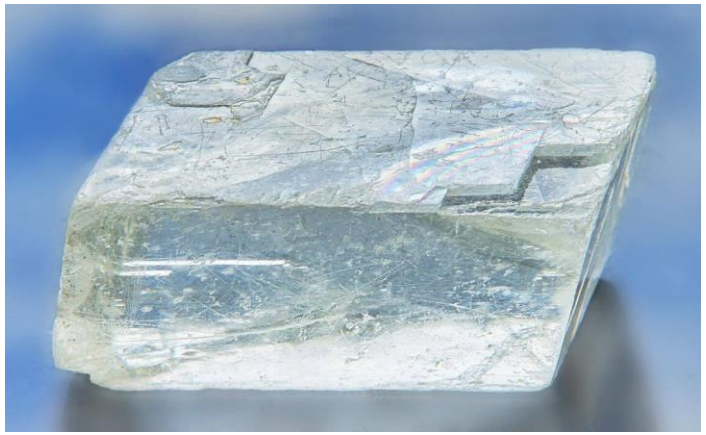
Formula: $CaCO_3$

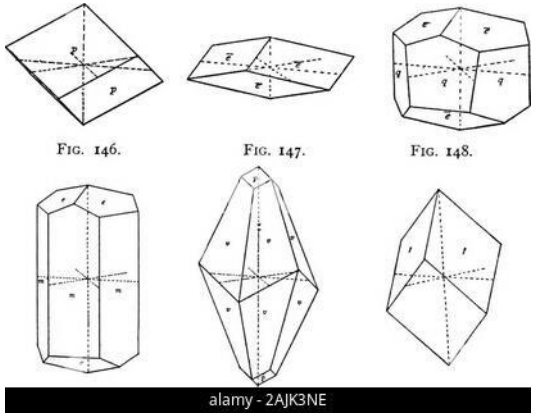
Color: White, Yellow, Red, Orange, Blue, Green, Brown, Gray

Lustre: Vitreous, Resinous, Waxy, Pearly

Hardness: 3

Crystal System: trigonal





Mineral identification sheet (4/4)

Halite

Formula: NaCl

Color: Colorless, whitish, purple, red, yellow, blue

Lustre: vitreous

Hardness: 2.5

Crystal System: cubic-isometric



Mica

Formula: $K(Mg,Fe)_2(Al,Fe,Mg,Ti)_2Si_3O_{10}(OH)_2$ (biotite)

$KAl_2(AlSi_3O_{10}(OH)_2$ (muscovite)

Color: dark green/brown, black (biotite); White to colorless, silvery-white, and tinged various colors by impurities (muscovite)

Lustre: vitreous

Hardness: 2.5-3

Crystal System: monoclinic

